

Migration to the canonical workable-items binary

Revision: 1 **Last modified:** 2026-05-31T00:00:00Z **Description:** Plan to migrate Lava ticket tracking from the bespoke tools/lava-tickets/ system (DB: docs/tickets/tickets.db) to the canonical workable-items Go binary shipped in the constitution submodule (constitution/scripts/workable-items/), per Constitution §11.4.93 + §11.4.74 catalogue-first. Tracked as LVA-4 ("LVA workable-items ticket system").

0. Executive summary (read first)

- **The LVA-key question is ANSWERED and requires NO upstream change.** The canonical add takes --prefix (default WIT; crud.go addPrefixDefault = "WIT"). --prefix LVA auto-allocates LVA-NNN (format "%s-%03d", crud.go nextID). --id LVA-N stores LVA-N **verbatim** (no padding). **CONFIRMED at runtime:** add --id LVA-7 → stored LVA-7; add --prefix LVA (no id) → stored LVA-008 (continues from max). To preserve the existing LVA-1..LVA-7 keys, pass --id LVA-N on each add. No constitution-submodule change is needed for LVA keys.
- **Build + run CONFIRMED.** Built from inside the module dir: (cd constitution/scripts/workable-items && CGO_ENABLED=1 go build -o /tmp/wi-canon ./cmd/workable-items) → **EXIT 0**, 7.4 MB binary. --help ran EXIT 0 (full output in §6). **go build from the repo ROOT FAILS** (go: cannot find main module — no root go.mod); you **MUST** cd into the module dir first. Self-contained go.mod (go 1.21, dep mattn/go-sqlite3 v1.14.22) + committed go.sum; schema is //go:embed schema_embed.sql. **CGO_ENABLED=1 REQUIRED** (cgo sqlite driver; CGO_ENABLED=0 fails).
- **End-to-end CONFIRMED** on a throwaway DB: close LVA-7 --status fixed moved it to Fixed with status Fixed (→ Fixed.md); validate → "OK"; sync db-to-md + diff → "DB and Markdown are in sync" EXIT 0.
- **The 7 LVA items CAN be represented**, but the schemas differ substantially (§3). Hard parts: (a) LVA priority P0–P3 vs canonical free-text severity; (b) LVA Types Incident/Debt/Chore are NOT in the canonical closed set {Bug,Feature,Task} — and add does NOT reject them, it **silently coerces unknown types to Task** (CONFIRMED: add Incident and add Debt both stored Task; parse.go normalizeType default: return "Task"). So to store

LVA-5 (Incident) as Bug, you MUST pass add Bug ..., never add Incident (c) LVA's provenance columns (source, source_ref, reopened_*, fix_commit_sha, validation_test, challenge_test, closure_log) have no canonical column.

§11.4.74 Catalogue-Check for the migration commit:

Catalogue-Check: reuse HelixDevelopment/HelixConstitution@<pin-sha> scripts/workable-items — canonical binary covers the core lava-tickets functionality; bespoke system retired.

1. Build + invoke from the Lava root

```
constitution/scripts/workable-items/  
├─ go.mod / go.sum      module github.com/HelixDevelopment/  
HelixConstitution/scripts/workable-items; go 1.21; mattn/go-  
sqlite3 v1.14.22  
├─ schema.sql           authoritative DDL (items + item_history +  
obsolete_details + operator_block_details + firebase_metadata +  
doc_segments + meta)  
└─ cmd/workable-items/  main.go crud.go db.go parse.go sync.go  
report.go schema_embed.sql
```

Build (resource-capped per §11.4.82 / §6.T.2) — MUST cd into the module dir:

```
( cd constitution/scripts/workable-items && \  
  CGO_ENABLED=1 GOMAXPROCS=2 nice -n 19 go build -o bin/workable-  
  items ./cmd/workable-items )
```

The submodule's .gitignore ignores bin/, *.db, so a built binary is not committed into the submodule.

Invoke (every subcommand requires --db; positional+flag order is free per partitionArgs):

```
WI=constitution/scripts/workable-items/bin/workable-items  
DB=docs/workable_items.db  
$WI add <Bug|Feature|Task> <severity> --db $DB --id LVA-N --title  
  "T" --description "D..."  
$WI close LVA-N --db $DB --status <fixed|implemented|completed|  
  obsolete> --evidence <path>  
$WI sync db-to-md --db $DB --out-issues docs/Issues.md --out-  
  fixed docs/Fixed.md  
$WI sync md-to-db --db $DB --issues docs/Issues.md --fixed docs/  
  Fixed.md  
$WI diff --db $DB --issues docs/Issues.md --fixed docs/  
  Fixed.md # exit 1 on divergence (CI gate)  
$WI validate --db $DB  
$WI report --db $DB [--by-type|--by-status|--by-severity|--by-  
  assigned|--by-creator|--obsolete-audit]
```

There is **no update and no reopen subcommand** — add always creates Queued; close is the only status-transition. Non-Queued open statuses need a DB-direct UPDATE (§4 Step 2c).

2. DB path + doc path

- **DB path = docs/workable_items.db** (§11.4.95 amends §11.4.93's dotfile to this tracked, non-dotfile path). Not hardcoded — pass --db; openDB os.MkdirAlls the parent dir.
 - **§11.4.95 — the DB is TRACKED in git, NEVER gitignored.** git add docs/workable_items.db. The .db-wal/.db-shm sidecars MUST be gitignored; run PRAGMA wal_checkpoint(TRUNCATE) before staging (schema sets journal_mode = WAL).
 - **Doc paths** are passed explicitly (no built-in default). Recommended: emit **docs/Issues.md + docs/Fixed.md** at the docs root (the §11.4.48 carve-out treats the canonical trackers as docs/-root constants). Lava keeps them under docs/tickets/ today; migrating to docs/ root aligns with the constitution. This plan assumes docs/ root.
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3. Schema comparison — the load-bearing differences

Real lava-tickets schema (docs/tickets/tickets.db, confirmed via .schema)

```
tickets(seq INTEGER PK AUTOINCREMENT, id TEXT UNIQUE ('LVA-<seq>'
via trigger),
  title, body, type->item_type, status->item_status,
closure_status->closure_status,
  priority CHECK('P0'..'P3'),
  source CHECK('operator-report','crashlytics','self-
discovered','bluff-hunt','code-review','user-report','automated-
gate'),
  source_ref, reopened_{why,who,when,incident}, fix_commit_sha,
validation_test,
  challenge_test, closure_log, operator_blocked_details,
duplicate_of,
  created_at, updated_at, closed_at)
+ lookup tables item_type/item_status/closure_status/
closure_status_for_type + triggers + views
```

LVA closed sets (confirmed):

- item_status: Blocked | Closed | InProgress | InReview | Open | OperatorBlocked | Reopened

- item_type: Bug | Chore | Debt | Feature | Incident | Task
- closure_status: Closed | Completed | Duplicate | Fixed | Implemented | NotReproducible | Resolved | WontFix

LVA id is LVA-<seq> (bare integer, NOT zero-padded).

Canonical schema (constitution/scripts/workable-items/schema.sql)

```
items(atm_id, type CHECK('Bug','Feature','Task'), status CHECK(10-
value closed set),
  severity, title, description, forensic_anchor, closure_criteria,
composes_with,
  created_by, assigned_to, current_location
CHECK('Issues','Fixed'), body_md,
  created_at, last_modified, PRIMARY KEY(atm_id,
current_location))
+ item_history / obsolete_details / operator_block_details /
firebase_metadata / doc_segments / meta
```

Canonical status closed set: Queued | In progress | Ready for testing | In testing | Reopened | Operator-blocked | Fixed (→ Fixed.md) | Implemented (→ Fixed.md) | Completed (→ Fixed.md) | Obsolete (→ Fixed.md).

Field mapping

lava-tickets	canonical	maps cleanly?
id (LVA-N)	items.atm_id	YES — pass --id LVA (verbatim, CONFIRMED). Composite PK (atm_id,current_location) allows same id in both trackers; each item has exactly one location, no conflict.
seq	—	DROPPED. No seq in canonical. Future auto-ids zero-pad (LVA-008) while legacy are bare (LVA-7); atoiSafe parses both so monotonicity holds. Cosmetic only.
type	items.type	TRANSFORM REQUIRED. Canonical only stores {Bug,Feature,Task} and add silently coerces

lava-tickets	canonical	maps cleanly?
		anything else to Task (CONFIRMED). Map the add command: Incident → Bug (pass Bug), Debt → Task, Chore → Task. Bug/Feature/Task pass through. LVA-2=Debt→Task, LVA-5=Incident→B TRANSFORM REQUIRED. Open→Queued, InProgress→In progress, OperatorBlocked→Operator blocked, Closed→ type-aware terminal (§11.4.33), Reopened→Reopened, Blocked/InReview→ nearest (In progress) YES via §11.4.33 — terminal status word the closure status (`close --status fixed store the P-level literal as severity (free text). No information lost. YES YES — §11.4.91 floor (≥6 words OR ≥40 chars) enforced at actual LVA bodies satisfy it. terminal status → Fixed else Issues. no column. Fold source_ref into description as a **Source:** line so the audit link survives. none of the 7 is Reopened now → no port. close records ONE evidence path; use the richest (closure_log or validation_test).
status	items.status	
closure_status	folded into items.status	
priority (P0-P3)	items.severity	
title	items.title	
body	items.description	
(derived Issues/Fixed)	items.current_location	
source,source_ref	—	
reopened_*	item_history (Reopened,by,reason,evidence_path)	
fix_commit_sha/ validation_test/ challenge_test/ closure_log	close --evidence → item_history.evidence_path	

lava-tickets	canonical	maps cleanly?
operator_blocked_details	operator_block_details table	structurally yes, but subcommand popular — DB-direct INSERT (§4 Step 2c). LVA-5 is OperatorBlocked.

The 7 real LVA items (confirmed from tickets.db)

id	LVA type	LVA status / closure	priority	→ canonical add type	→ canonical status	→ location
LVA-1	Bug	Closed / Fixed	P1	Bug	Fixed (→ Fixed.md)	Fixed
LVA-2	Debt	Closed / Closed	P1	Task	Completed (→ Fixed.md)	Fixed
LVA-3	Task	InProgress	P1	Task	In progress	Issues
LVA-4	Feature	InProgress	P1	Feature	In progress	Issues
LVA-5	Incident	OperatorBlocked	P0	Bug (NOT Incident — would coerce to Task)	Operator-blocked	Issues
LVA-6	Task	Open	P2	Task	Queued	Issues
LVA-7	Task	Open	P2	Task	Queued	Issues

Titles/bodies: quote verbatim from
 sqlite3 docs/tickets/tickets.db "SELECT id,title,body FROM tickets ORDER BY seq" — do NOT paraphrase (§11.4.6).

4. Step-by-step migration (main-agent-executable)

CWD = repo root. WI=constitution/scripts/workable-items/
 bin/workable-items, DB=docs/workable_items.db.

Step 1 — build + capture evidence

```
( cd constitution/scripts/workable-items && CGO_ENABLED=1
  GOMAXPROCS=2 nice -n 19 go build -o bin/workable-items ./
  cmd/workable-items )
"$WI" --help      # capture as §11.4.5 evidence (EXIT 0 confirmed)
```

Step 2 — seed 7 items, preserving exact keys

Pull --title/--description from the DB for each. Use the canonical add type per the §3 table (Incident→Bug, Debt→Task). Severity = the P-level.

```
rm -f "$DB" "$DB-wal" "$DB-shm"
"$WI" add Bug      P1 --db "$DB" --id LVA-1 --title "<LVA-1
  title>" --description "<LVA-1 body>"
"$WI" add Task     P1 --db "$DB" --id LVA-2 --title "<LVA-2
  title>" --description "<LVA-2 body>"
"$WI" add Task     P1 --db "$DB" --id LVA-3 --title "<LVA-3
  title>" --description "<LVA-3 body>"
"$WI" add Feature  P1 --db "$DB" --id LVA-4 --title "<LVA-4
  title>" --description "<LVA-4 body>"
"$WI" add Bug      P0 --db "$DB" --id LVA-5 --title "<LVA-5
  title>" --description "<LVA-5 body>" # Incident→Bug
"$WI" add Task     P2 --db "$DB" --id LVA-6 --title "<LVA-6
  title>" --description "<LVA-6 body>"
"$WI" add Task     P2 --db "$DB" --id LVA-7 --title "<LVA-7
  title>" --description "<LVA-7 body>"
```

Step 2b — close the two terminal items (LVA-1, LVA-2)

```
"$WI" close LVA-1 --db "$DB" --status fixed \
  --evidence feature/credentials/src/test/kotlin/lava/feature/
  credentials/CredentialsViewModelTest.kt
"$WI" close LVA-2 --db "$DB" --status completed \
  --evidence .lava-ci-evidence/sixth-law-incidents/2026-05-13-
  emulator-container-darwin-arm64-gap.json
```

Both paths are the genuine evidence already in the LVA rows (honest, non-bluff).

Step 2c — non-Queued open statuses (LVA-3, LVA-4 In progress; LVA-5 Operator-blocked)

add only produces Queued; no update subcommand. DB-direct (recommended now):

```
sqlite3 "$DB" "UPDATE items SET status='In progress',
  body_md=replace(body_md,'**Status:** Queued','**Status:** In
  progress');"
WHERE atm_id IN ('LVA-3','LVA-4');"
```

```
sqlite3 "$DB" "UPDATE items SET status='Operator-blocked',
  body_md=replace(body_md,'**Status:** Queued','**Status:**
  Operator-blocked')
  WHERE atm_id='LVA-5';"
sqlite3 "$DB" "INSERT INTO
  operator_block_details(atm_id,what,why_exhausted_alternatives,unblock_command)
  VALUES('LVA-5','<what from tickets.db
  operator_blocked_details>','<why>','<unblock>','Operator');"
```

The body_md replace() keeps the §11.4.93 round-trip green.
(Alternative: extend the canonical binary upstream with update/
block subcommands per §11.4.74 — OWED, not blocking.)

Step 3 — generate doc set + verify

```
sqlite3 "$DB" "PRAGMA wal_checkpoint(TRUNCATE);"
"$WI" sync db-to-md --db "$DB" --out-issues docs/Issues.md --out-
  fixed docs/Fixed.md
"$WI" validate --db "$DB" # expect "OK — 7 items"
"$WI" diff --db "$DB" --issues docs/Issues.md --fixed docs/
  Fixed.md # expect "in sync", exit 0
"$WI" report --db "$DB" --by-status
```

Binary generates **Issues.md + Fixed.md only**. Summaries
(Issues_Summary.md/Fixed_Summary.md) + HTML/PDF (§11.4.12/
§11.4.53/§11.4.65) are NOT produced by it — tools/lava-tickets/
export.go/render.go produced them today. After retiring that,
regenerate via the project export pipeline OR the §11.4.106 Docs
Chain engine. Do NOT leave stale docs/tickets/*_Summary.* / docs/
tickets/export/* behind.

Step 4 — retire the bespoke system (main agent does git rm)

- tools/lava-tickets/ (whole dir).
- docs/tickets/tickets.db → replaced by docs/workable_items.db.
- docs/tickets/schema.sql → replaced by the canonical embedded schema.
- docs/tickets/{Issues,Fixed,Issues_Summary,Fixed_Summary}.md + docs/tickets/export/* → replaced by docs/Issues.md/docs/Fixed.md + regenerated summaries/exports.
- Keep docs/tickets/DESIGN.md + docs/tickets/BUILD-EVIDENCE.md as forensic anchors (or move under docs/tickets/retired/). Any path used as --evidence MUST remain reachable.

Replace the CM-LVA-TICKETS-SYNC gate. UNCONFIRMED: its exact wiring location — the grep for CM-LVA-TICKETS-SYNC/lava-tickets in scripts/check-constitution.sh + .github/pre-push did not return output this session (session I/O degradation), so the main agent MUST grep to locate it before editing. Replace its body with:

- workable-items validate --db docs/workable_items.db

- `workable-items diff --db docs/workable_items.db --issues docs/Issues.md --fixed docs/Fixed.md (exit 1 = §11.4.93 CM-WORKABLE-ITEMS-MD-DB-IN-SYNC)`
- `git ls-files docs/workable_items.db non-empty = §11.4.95 CM-WORKABLE-ITEMS-DB-TRACKED`
- `gate` builds `bin/workable-items` if absent, or go `run ./cmd/workable-items` from the module dir.

Step 5 — governance + gitignore + CONTINUATION (SAME commit, §6.S/§12.10)

- **.gitignore:** add `docs/workable_items.db-wal + docs/workable_items.db-shm`. Do NOT ignore `docs/workable_items.db` (§11.4.95). Remove dead `docs/tickets/` lines.
- **CLAUDE.md §6.AF:** rewrite to point at the canonical binary (`constitution/scripts/workable-items/`), DB `docs/workable_items.db`, docs `docs/Issues.md/docs/Fixed.md`, the LVA-prefix convention (`--id LVA-N` to preserve, `--prefix LVA` for new), the Incident→Bug / Debt→Task / Chore→Task type-mapping decision, and the new gate. Add Catalogue-Check: reuse ... + Classification: project-specific.
- **docs/CONTINUATION.md:** record LVA-4 progress, new DB+doc paths, retired tools/lava-tickets/, OWED follow-ups (canonical update/reopen/block subcommand extension; summary+export pipeline standardisation; the Incident/Debt/Chore type-coercion-to-Task caveat; source/source_ref folded into description).

Step 6 — Catalogue-Check + commit (main agent)

Catalogue-Check: reuse HelixDevelopment/HelixConstitution@<pin-sha> `scripts/workable-items`
 Classification: project-specific (Lava consumes the canonical binary; LVA-prefix + Incident→Bug/Debt→Task/Chore→Task mapping is Lava-specific)

5. Main-agent-doable vs operator/upstream-blocked

Item	Status
Build canonical binary	DONE/feasible (CONFIRMED EXIT 0 from module dir; root build fails — cd in first).
Seed 7 items with exact keys (<code>--id LVA-N</code>)	Main-agent-doable (CONFIRMED <code>--id verbatim + --prefix auto-alloc</code>).

Item	Status
Type transform (Incident→Bug, Debt/ Chore→Task)	Main-agent-doable (mandatory: add Incident/add Debt silently coerce to Task — CONFIRMED — so pass the canonical type explicitly).
Status transform + close LVA-1/LVA-2	Main-agent-doable (CONFIRMED close→Fixed + validate OK).
LVA-3/4 In progress, LVA-5 Operator-blocked + operator_block_details	Main-agent-doable via DB- direct UPDATE + body_md replace + INSERT (no update/block subcommand).
Generate Issues.md/ Fixed.md + validate + diff	Main-agent-doable (CONFIRMED round-trip in sync).
Retire tools/lava-tickets/ + replace gate + update CLAUDE/.gitignore/ CONTINUATION	Main-agent-doable. Grep CM- LVA-TICKETS-SYNC first to locate gate wiring (UNCONFIRMED: location this session).
Summary docs + HTML/PDF exports	Main-agent-doable but depends on pandoc/weasyprint + project export pipeline (binary does NOT emit them).
update/reopen/block upstream extensions	OWED upstream-extend (§11.4.74), NOT blocking. Requires §11.4.26 7-step constitution-submodule pipeline + push to its 4 upstreams. File as a future LVA item. --prefix default change is NOT needed.

6. Captured evidence (this session, from this checkout)

go build from inside constitution/scripts/workable-items with
CGO_ENABLED=1 → **EXIT 0**, 7.4 MB binary. go build from repo root →
FAILS go: cannot find main module.

--help (EXIT 0, verbatim):

workable-items - §11.4.93 SQLite-SSoT for workable items

Usage:

workable-items <subcommand> [args...]

Subcommands:

```
sync md-to-db --db <p> [--issues <p>] [--fixed <p>] Parse
trackers, upsert DB.
sync db-to-md --db <p> [--out-issues <p>] [--out-fixed <p>]
Regenerate trackers from DB.
diff --db <p> [--issues <p>] [--fixed <p>] Show DB vs
Markdown divergence.
validate --db <p> Closed-set
+ §11.4.91 invariants.
add <type> <severity> --db <p> --title <T> --description <D> [--
id <id>] [--prefix <P>] [--created-by <h>] [--assigned-to <h>]
Create a
new Queued item in Issues.
close <atm-id> --db <p> --status <fixed|implemented|completed|
obsolete> --evidence <p>
Atomic
Issues→Fixed closure (§11.4.19).
report --db <p> [--by-type|--by-status|--by-severity|--by-
assigned|--by-creator|--obsolete-audit]
Read-only
grouped tally / §11.4.90 audit.
```

Canonical authority: Constitution.md §11.4.93.

Runtime behaviour proofs (throwaway DB):

- add Bug P0 --id LVA-7 ... → add: created LVA-7 (Bug, status=Queued); stored atm_id = LVA-7.
- add Task P2 --prefix LVA ... (no id) → add: created LVA-008 (Task, status=Queued); stored atm_id = LVA-008.
- add Incident ... → (Task, ...) and add Debt ... → (Task, ...) — **unknown types silently coerced to Task.**
- close LVA-7 --status fixed --evidence <p> → close: moved LVA-7 Issues→Fixed (status=Fixed (→ Fixed.md)).
- validate → validate: OK – 2 items, all invariants satisfied.
- sync db-to-md + diff → diff: DB and Markdown are in sync (EXIT 0).

7. The LVA-key answer (definitive)

Configurable — NO upstream change required. Source (crud.go): const addPrefixDefault = "WIT"; nextID returns fmt.Sprintf("%s-%03d", prefix, max+1); explicit --id stored as strings.TrimSpace(*explicitID) verbatim. Confirmed at runtime (§6). Preserve LVA-1..LVA-7 via --id LVA-N; new tickets via --prefix LVA (zero-padded LVA-NNN). Neither path needs a constitution-submodule edit.